

Problem 15.1.16

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(20 + 30i)^n}{n!}$$

Solution

Ratio Test:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(20 + 30i)^{n+1}}{(n+1)!} \cdot \frac{n!}{(20 + 30i)^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{20 + 30i}{n+1} \right| \\ &= \lim_{n \rightarrow \infty} \frac{\sqrt{1300}}{n+1} \\ &= 0 < 1, \text{ therefore the series converges} \end{aligned}$$

Problem 15.1.17

Is the series convergent or divergent? Give a reason.

$$\sum_{n=2}^{\infty} \frac{(-i)^n}{\ln n}$$

Solution

The numerator is bounded cyclically so only the denominator affects the series convergence.

$$\frac{1}{\ln n} > \frac{1}{n} \quad \forall n \in \mathbb{Z} \geq 2$$

Therefore, the series diverges by direct comparison to the divergent harmonic series.

Problem 15.1.18

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} n^2 \left(\frac{i}{4} \right)^n$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2 \left(\frac{i}{4}\right)^{n+1}}{n^2 \left(\frac{i}{4}\right)^n} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{n^2} \cdot \frac{i}{4} \right| \\
&= \frac{1}{4} \cdot \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} \\
&= \frac{1}{4} \cdot 1 \\
&= \frac{1}{4} < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.19

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{i^n}{n^2 - i}$$

Solution

$$|a_n| = \left| \frac{i^n}{n^2 - i} \right| = \frac{1}{\sqrt{n^4 + 1}}$$

$$|a_n| < \frac{1}{n^2} \quad \forall n \in \mathbb{Z} > 0$$

Therefore, the series converges by direct comparison to the convergent p-series $\sum_{n=0}^{\infty} \frac{1}{n^2}$.**Problem 15.1.20**

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{n+i}{3n^2+2i}$$

Solution

The real part diverges due to the harmonic series and therefore the series diverges.

$$\sum_{n=0}^{\infty} \frac{n}{3n^2} = \frac{1}{3} \sum_{n=0}^{\infty} \frac{1}{n}$$

Problem 15.1.21

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(\pi + \pi i)^{2n+1}}{(2n+1)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(\pi + \pi i)^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{(\pi + \pi i)^{2n+1}} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(\pi + \pi i)^2}{(2n+3)(2n+2)} \right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.22

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

Solution

$$\frac{1}{\sqrt{n}} > \frac{1}{n} \quad \forall n \in \mathbb{Z} > 1$$

Therefore, the series diverges by direct comparison with the harmonic series.

Problem 15.1.23

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(-1)^n (1+i)^{2n}}{(2n)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (1+i)^{2(n+1)}}{(2(n+1))!} \cdot \frac{(2n)!}{(-1)^n (1+i)^{2n}} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{-(1+i)^2}{(2n+2)(2n+1)} \right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.24

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} \frac{(3i)^n n!}{n^n}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(3i)^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{(3i)^n n!} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(3i)^{n+1}}{(3i)^n} \cdot \frac{(n+1)!}{n!} \cdot \frac{n^n}{(n+1)^{n+1}} \right| \\
&= \lim_{n \rightarrow \infty} \left| 3i \cdot (n+1) \cdot \frac{n^n}{(n+1)^n(n+1)} \right| \\
&= 3 \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1} \right)^n \right| \\
&= 3 \lim_{n \rightarrow \infty} \left| \frac{1}{\left(\frac{n+1}{n} \right)^n} \right| \\
&= 3 \lim_{n \rightarrow \infty} \left| \frac{1}{\left(1 + \frac{1}{n} \right)^n} \right| \\
&= 3 \cdot \frac{1}{e} \\
&= \frac{3}{e} > 1, \text{ therefore the series diverges}
\end{aligned}$$

Problem 15.2.7

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(z - \frac{1}{2} \right)^{2n}$$

Solution

$$\begin{aligned}
R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{(2n)!} \cdot \frac{(2n+2)!}{(-1)^{n+1}} \right| \\
&= \lim_{n \rightarrow \infty} |(2n+2)(2n+1)| \\
&= \infty
\end{aligned}$$

The center of convergence is at $z_0 = \frac{1}{2}$ with radius $R = \infty$.**Problem 15.2.9**

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{n(n-1)}{3^n} (z-i)^{2n}$$

SolutionRearranging the power series to the classic form by substituting $n = 2n$,

$$\sum_{n=0}^{\infty} \frac{\frac{n}{2}(\frac{n}{2}-1)}{3^{\frac{n}{2}}}(z-i)^n = \frac{1}{4} \sum_{n=0}^{\infty} \frac{n(n-2)}{3^{\frac{n}{2}}}(z-i)^n$$

$$\begin{aligned} R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n(n-2)}{3^{\frac{n}{2}}} \cdot \frac{3^{\frac{n+1}{2}}}{(n+1)(n-1)} \right| \\ &= \sqrt{3} \lim_{n \rightarrow \infty} \left| \frac{n(n-2)}{(n+1)(n-1)} \right| \\ &= \sqrt{3} \end{aligned}$$

The center of convergence is at $z_0 = i$ with radius $R = \sqrt{3}$.

Problem 15.2.13

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} 16^n (z+i)^{4n}$$

Solution

Rearranging the power series to the classic form by substituting $n = 4n$,

$$\sum_{n=0}^{\infty} 16^{\frac{n}{4}} (z+i)^n$$

$$\begin{aligned} R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{16^{\frac{n}{4}}}{16^{\frac{n+1}{4}}} \right| \\ &= \lim_{n \rightarrow \infty} \left| 16^{-\frac{1}{4}} \right| \\ &= \frac{1}{2} \end{aligned}$$

The center of convergence is at $z_0 = -i$ with radius $R = \frac{1}{2}$.

Problem 15.2.15

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2} (z-2i)^n$$

Solution

$$\begin{aligned}
R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(2n)!}{4^n (n!)^2} \cdot \frac{4^{n+1} ((n+1)!)^2}{(2n+2)!} \right| \\
&= \lim_{n \rightarrow \infty} \left| 4 \cdot \frac{(2n)!}{(2n+2)!} \cdot \frac{((n+1)!)^2}{(n!)^2} \right| \\
&= \lim_{n \rightarrow \infty} \left| 4 \cdot \frac{1}{(2n+2)(2n+1)} \cdot (n+1)^2 \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{2n+2}{2n+1} \right| \\
&= 1
\end{aligned}$$

The center of convergence is at $z_0 = 2i$ with radius $R = 1$.

Problem 15.2.17

Find the center and the radius of convergence.

$$\sum_{n=1}^{\infty} \frac{2^n}{n(n+1)} z^{2n+1}$$

Solution

$$\begin{aligned}
L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}}{(n+1)(n+2)} z^{2n+3} \cdot \frac{n(n+1)}{2^n} z^{-(2n+1)} \right| \\
&= \lim_{n \rightarrow \infty} \left| 2z^2 \cdot \frac{n}{n+1} \right| \\
&= 2|z|^2
\end{aligned}$$

$$\begin{aligned}
L &< 1 \\
2|z|^2 &< 1 \\
|z|^2 &< \frac{1}{2} \\
|z| &< \frac{1}{\sqrt{2}}
\end{aligned}$$

The center of convergence is at $z_0 = 0$ with radius $R = \frac{1}{\sqrt{2}}$.

Problem 15.3.5

Find the radius of convergence in two ways:

- Directly by the Cauchy–Hadamard formula in Sec. 15.2
- From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} (z-2i)^n$$

Solution

Cauchy-Hadamard

$$\begin{aligned} R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n(n-1)}{2^n} \cdot \frac{2^{n+1}}{(n+1)n} \right| \\ &= 2 \cdot \lim_{n \rightarrow \infty} \left| \frac{n-1}{n+1} \right| \\ &= 2 \end{aligned}$$

Theorem 3: Differentiation

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{z-2i}{2} \right)^n = \sum_{n=0}^{\infty} \frac{1}{2^n} (z-2i)^n$$

The geometric series $f(z)$ converges if

$$\frac{z-2i}{2} < 1 \implies R = 2$$

Differentiating,

$$\begin{aligned} f'(z) &= \sum_{n=1}^{\infty} \frac{n}{2^n} (z-2i)^{n-1} \\ f''(z) &= \sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} (z-2i)^{n-2} \end{aligned}$$

The original series is equivalent to $f''(z)(z-2i)^2$ and therefore also converges with $R = 2$

Problem 15.3.7

Find the radius of convergence in two ways:

- Directly by the Cauchy-Hadamard formula in Sec. 15.2
- From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{n}{3^n} (z+2i)^{2n}$$

Solution**Cauchy-Hadamard**

$$\begin{aligned}
L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{3^{n+1}}{n+1} (z+2i)^{2n+2} \cdot \frac{n}{3^n} (z+2i)^{-2n} \right| \\
&= \frac{1}{3} \cdot \lim_{n \rightarrow \infty} \left| (z+2i)^2 \cdot \frac{n+1}{n} \right| \\
&= \frac{1}{3} |z+2i|^2
\end{aligned}$$

$$\begin{aligned}
\frac{1}{3} |z+2i|^2 &< 1 \\
|z+2i|^2 &< 3 \\
|z+2i| &< \sqrt{3}
\end{aligned}$$

Therefore $R = \sqrt{3}$

Theorem 3: Differentiation

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{(z+2i)^2}{3} \right)^n = \sum_{n=0}^{\infty} \frac{1}{3^n} (z+2i)^{2n}$$

The geometric series $f(z)$ converges if

$$\frac{(z+2i)^2}{3} < 1 \implies R = \sqrt{3}$$

Differentiating,

$$f'(z) = \sum_{n=1}^{\infty} \frac{2n}{3^n} (z+2i)^{2n-1}$$

The original series is equivalent to $\frac{1}{2}(z+2i)f'(z)$ and therefore also converges with $R = \sqrt{3}$

Problem 15.3.9

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n(n+1)(n+2)} z^{2n}$$

Solution**Cauchy-Hadamard**

$$\begin{aligned}
L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\
&= \lim_{n \rightarrow \infty} \left| \frac{(-2)^{n+1}}{(n+1)(n+2)(n+3)} z^{2n+2} \cdot \frac{n(n+1)(n+2)}{(-2)^n} z^{-2n} \right| \\
&= 2|z|^2 \cdot \lim_{n \rightarrow \infty} \left| \frac{n(n+1)(n+2)}{(n+1)(n+2)(n+3)} \right| \\
&= 2|z|^2
\end{aligned}$$

$$\begin{aligned}
L &< 1 \\
2|z|^2 &< 1 \\
|z|^2 &< \frac{1}{2} \\
|z| &< \frac{1}{\sqrt{2}}
\end{aligned}$$

Therefore $R = \frac{1}{\sqrt{2}}$

Theorem 4: Integration

$$f(z) = \sum_{n=0}^{\infty} (-2z^2)^n$$

The geometric series $f(z)$ converges if

$$|-2z^2| < 1 \implies R = \frac{1}{\sqrt{2}}$$

Integrating 3 times introduces the factors $\frac{1}{n+1}$, $\frac{1}{n+2}$, and $\frac{1}{n+3}$ which is similar to the original series and it therefore also converges with $R = \frac{1}{\sqrt{2}}$

Problem 15.3.11

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy-Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{3^n n(n+1)}{7^n} (z+2)^{2n}$$

Solution**Problem 15.3.13**

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy-Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=0}^{\infty} \left[\binom{n+k}{k} \right]^{-1} z^{n+k}$$

Solution